

CII DOCUMENTATION

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WHAT IS CII :

- It is Carbon Intensity Indicator
- It evaluates the transport efficiency of a vessel
- The CII (Carbon Intensity Indicator) is a measure for a ship's energy efficiency and is given in grams of CO₂ emitted per cargo-carrying capacity and nautical mile. ie, how much carbon emitted to the atmosphere for a transport work.
- Carbon emissions intensity is important because it allows us to measure progress in reducing carbon emissions, even as total emissions may continue to increase. By reducing emissions intensity, countries and companies can become more environmentally sustainable and reduce their impact on climate change.
- In 2023 the International Maritime Organization is expected to implement new initiatives to cut the shipping industry's carbon footprint. The new greenhouse gas (GHG) strategy will put increasing pressure on all industry actors and require ship owners to report their sustainable operations annually. The Carbon Intensity Indicator (CII), effective from January 2023, will indicate the vessel's operational efficiency.

CII RATING SYSTEM :

- The CII provides a rating system to grade ships based on how much air pollution they are responsible for
- Ratings from A to E are classified as:
 - A – Major superior performance level
 - B – Minor superior performance level
 - C – Moderate performance level
 - D – Minor Inferior performance level
 - E – Inferior performance level

The efficiency of the vessel decreases as we go from A to E. Within each grade, we have a range of CII rating values. For instance, a vessel may be on the better or worse side of the C rating. Vessels graded D and E for three years must improve their score.

All vessels must aspire to be at least C-rated. The rating will be recorded in the Ship Energy Efficiency Management Plan (SEEMP).

Any vessel that is rated D or E for three years in a row will have to submit a corrective action plan that explains the efforts that will be taken to achieve an acceptable rating (C or above). This plan must be approved by the flag state administration or any Recognized Organization (RO).

CII CALCULATION METHODS :

To understand where our ship lies, we must first understand and calculate three types of CII values – **Reference CII**, **Required CII** and **Attained CII**.

● **REFERENCE CII (CII_{ref}):**

The operational carbon intensity performance of ship types in the year 2019 is taken as the reference (**RESOLUTION MEPC.337(76), Annex 11**).

$$CII_{ref} = a.Capacity^{-c}$$

Parameters for determining the ship type specific reference line

Ship type		Capacity	<i>a</i>	<i>c</i>
Bulk carrier	279,000 DWT and above	279,000	4745	0.622
	less than 279,000 DWT	DWT	4745	0.622
Gas carrier	65,000 and above	DWT	14405E7	2.071
	less than 65,000 DWT	DWT	8104	0.639
Tanker		DWT	5247	0.610
Container ship		DWT	1984	0.489
General cargo ship	20,000 DWT and above	DWT	31948	0.792
	less than 20,000 DWT	DWT	588	0.3885
Refrigerated cargo carrier		DWT	4600	0.557
Combination carrier		DWT	40853	0.812
LNG carrier	100,000 DWT and above	DWT	9.827	0.000
	65,000 DWT and above, but less than 100,000 DWT	DWT	14479E10	2.673
	less than 65,000 DWT	65,000	14479E10	2.673
Ro-ro cargo ship (vehicle carrier)		GT	5739	0.631
Ro-ro cargo ship		DWT	10952	0.637
Ro-ro passenger ship		GT	7540	0.587
Cruise passenger ship		GT	930	0.383

● REQUIRED CII :

The value of the required CII depends on the reduction factor for the year. With every passing year, the CII limits will become stricter. In accordance with regulation 28 of MARPOL Annex VI, the required annual operational CII for a ship calculated as follows (**RESOLUTION MEPC.338(76), Annex 12**),

$$\text{Required CII} = \text{CII}_{\text{ref}} \times [(100 - Z)/100],$$

where **Z** is the reduction factor for the year.

Reduction factor (Z%) for the CII relative to the 2019 reference line

Year	Reduction factor relative to 2019
2023	5%*
2024	7%
2025	9%
2026	11%
2027	- **
2028	- **
2029	- **
2030	- **

* Z factors of 1% ,2% and 3% are set for the years of 2020, 2021, 2022 respectively.

**Z factors for the years of 2027 to 2030 to be further strengthened and developed taking into account the review of the short-term measure.

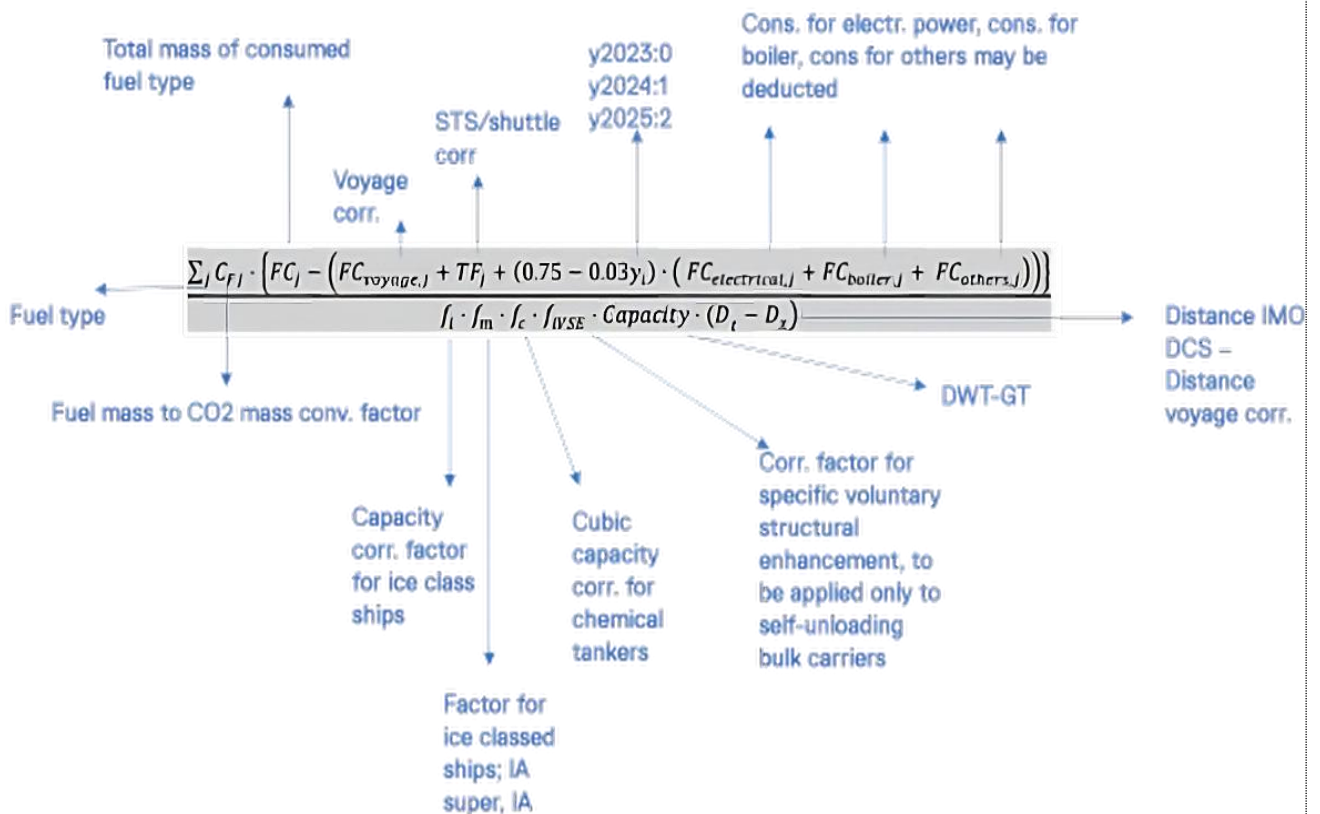
● ATTAINED CII :

The CII is derived from the Annual Emissions Ratio (AER): annual fuel consumption multiplied by the CO2 emissions factor, divided by the transport network (distance sailed by a ship multiplied by capacity). According to **RESOLUTION MEPC.336(76), Annex 10**, Attained CII is as follows,

$$\text{Attained CII} = \frac{\text{Mass of CO}_2 \text{ (M)}}{\text{Transport work (W)}}$$

$$\text{CII} = \frac{[\text{AnnualFuelConsumption}] \times [\text{CO}_2\text{emissionsfactor}]}{\text{TransportWork} : [\text{DistSailed}] \times [\text{Capacity}]}$$

Furthermore, there are several correction factors for specific ship types (see the image below). Corrections are applied for ships with ice class, refrigerated cargo, or heating or cooling systems for cargo or cargo handling systems (**RESOLUTION MEPC.355(78), Annex 17**). There are also additional exclusions, such as for shuttle tankers in ship-to-ship cargo operations, chemical tankers, and structurally enhanced bulk carriers.



ATTAINED CII CALCULATIONS FOR REFRIGERATED CONTAINERS :

The equation of ATTAINED CII will be ,

$$= \frac{\sum_j \cdot CF_j \{ FC_j - (0.75 - 0.03y_i) \cdot (FC_{\text{electrical},j}) \}}{\text{Capacity} \times \text{distance}}$$

1. **j** = Fuel Type
2. **FC_j** = Total mass of consumed Fuel
3. **CF_j** = Conversion factor (Fuel mass to CO₂ mass)

Type of fuel	Reference	Carbon content	C_F (t-CO ₂ /t-Fuel)
1. Diesel/Gas Oil	ISO 8217 Grades DMX through DMC	0.875	3.206000
2. Light Fuel Oil (LFO)	ISO 8217 Grades RMA through RMD	0.86	3.151040
3. Heavy Fuel Oil (HFO)	ISO 8217 Grades RME through RMK	0.85	3.114400
4. Liquified Petroleum Gas (LPG)	Propane	0.819	3.000000
	Butane	0.827	3.030000
5. Liquified Natural Gas (LNG)		0.75	2.750000

Note : In the case of containers, **FC_{voyage,j}**, **TF_j** , **FC_{boiler,j}**, **FC_{others,j}** will **not** be considered and **F_i = F_m = F_c = F_{i,vse} = 1**.

4. $FC_{\text{electrical},j}$ = Fuel consumption for production for electrical power.

For ships carrying refrigerated containers, the correction factor $FC_{\text{Electrical}}$ may be applied as follows:

1. For ships that **have the ability to monitor reefer electrical consumption**,

$$FC_{\text{electrical_reefer},j} = \text{Reefer kWh} \times \text{SFOC}$$

$FC_{\text{electrical_reefer},j}$ = Reefer fuel oil consumption

Reefer kWh = measured on the ship by the kWh meter counter on the ship

SFOC = weighted average SFOC of Auxiliary engines (a default value of **175 g/kWh** for **2 stroke** engines and **200 g/kWh** for **4 stroke** engines may be applied)

2. For ships that **do not** have the ability to **monitor reefer electrical consumption**,

$$FC_{\text{electrical_reefer},j} = Cx \cdot 24 \cdot \text{SFOC}_{\text{avg}} \cdot \left(\text{Reefer_days}_{\text{sea}} + \sum \text{Reefer_days}_{\text{port}} \right)$$

$Cx = 2.75 \text{ kW/h}$

Reefer_days_{sea} = number of reefer containers × number of days AT SEA

Reefer_days_{port} = represents the number of in-use reefer days while at port

SFOC_{avg} = weighted average SFOC of Auxiliary engines

$$\text{Reefer_days}_{\text{port}} = \frac{\text{No}_c \text{ Arrival} + \text{No}_c \text{ Departure}}{2} \times \text{Days}_{\text{port}}$$

1. $\text{Days}_{\text{port}}$ = Number of days in port

2. $\text{No}_c \text{ Arrival}$ = number of reefer containers on arrival.

3. $\text{No}_c \text{ Departure}$ = number of reefer containers on departure.

CII RATING :

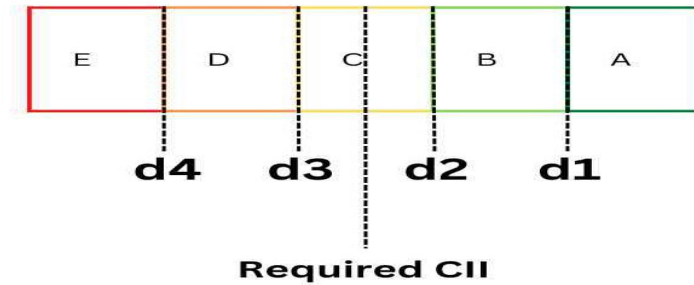
An operational energy efficiency performance rating should be annually assigned to each ship to which regulation 28 of MARPOL Annex IV applies, in a transparent and robust manner, based on the deviation of attained annual operational carbon intensity indicator (CII) of a ship from the required value.

$$\text{CII Ratio} = \frac{\text{Attained CII}}{\text{Required CII}}$$

dd vectors for determining the rating boundaries of ship types

Ship type		Capacity in CII calculation	<i>dd</i> vectors (after exponential transformation)			
			exp(d1)	exp(d2)	exp(d3)	exp(d4)
Bulk carrier		DWT	0.86	0.94	1.06	1.18
Gas carrier	65,000 DWT and above	DWT	0.81	0.91	1.12	1.44
	less than 65,000 DWT	DWT	0.85	0.95	1.06	1.25
Tanker		DWT	0.82	0.93	1.08	1.28
Container ship		DWT	0.83	0.94	1.07	1.19
General cargo ship		DWT	0.83	0.94	1.06	1.19
Refrigerated cargo carrier		DWT	0.78	0.91	1.07	1.20
Combination carrier		DWT	0.87	0.96	1.06	1.14
LNG carrier	100,000 DWT and above	DWT	0.89	0.98	1.06	1.13
	less than 100,000 DWT		0.78	0.92	1.10	1.37
Ro-ro cargo ship (vehicle carrier)		GT	0.86	0.94	1.06	1.16
Ro-ro cargo ship		DWT	0.66	0.90	1.11	1.37
Ro-ro passenger ship		GT	0.72	0.90	1.12	1.41
Cruise passenger ship		GT	0.87	0.95	1.06	1.16

The CII rating has five main grades: A, B, C, D and E. The efficiency of the vessel **decreases** as we go from **A to E**.



If ,

CII Ratio $\leq$ d1 -> A rating
d1 < CII Ratio $\leq$ d2 -> B rating
d2 < CII Ratio $\leq$ d3 -> C rating
d3 < CII Ratio $\leq$ d4 -> D rating
CII Ratio > d4 -> E rating

The efficiency of the vessel decreases as we go from A to E. Within each grade, we have a range of CII rating values. For instance, a vessel may be on the better or worse side of the C rating. Vessels graded D and E for three years must improve their score.